

Tugas Penerapan Transformasi Intensitas dan Ekualisasi Histogram

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Kode yang Digunakan

```
import cv2
import numpy as np

# BACA GAMBAR

image = cv2.imread("rb.png")

if image is None:
    print("Gambar tidak ditemukan!")
    exit()

height = image.shape[0]
width = image.shape[1]

# 1. GRAYSCALE MANUAL

# OpenCV membaca gambar dalam format BGR.
#
# Rumus grayscale :
#  $Gray = 0.299R + 0.587G + 0.114B$ 

gray = np.zeros((height, width), dtype=np.uint8)

for y in range(height):
    for x in range(width):

        B = int(image[y, x, 0])
        G = int(image[y, x, 1])
        R = int(image[y, x, 2])

        nilai_gray = 0.114 * B + 0.587 * G + 0.299 * R

        gray[y, x] = int(nilai_gray)

# 2. TRANSFORMASI INTENSITAS
```

```

# A. BRIGHTNESS – MENCERAHKAN GAMBAR

# Rumus:
#  $s = r + C$ 

C = 50

brightness = np.zeros((height, width), dtype=np.uint8)

for y in range(height):
    for x in range(width):

        r = int(gray[y, x])

        s = r + C

        if s > 255:
            s = 255

        brightness[y, x] = s


# B. DARKNESS – MENGGELAPKAN GAMBAR

# Rumus:
#  $s = r - C$ 

C = 50

darkness = np.zeros((height, width), dtype=np.uint8)

for y in range(height):
    for x in range(width):

        r = int(gray[y, x])

        s = r - C

        if s < 0:
            s = 0

        darkness[y, x] = s


# C. NEGATIVE IMAGE

# Rumus:
#  $s = 255 - r$ 

negative = np.zeros((height, width), dtype=np.uint8)

for y in range(height):
    for x in range(width):

        r = int(gray[y, x])

```

```

s = 255 - r

negative[y, x] = s

```

D. CONTRAST STRETCHING

```

# Tujuan :
# Memperlebar rentang intensitas gambar.
#
# Rumus :
#
#  $s = ((r - r_{min}) / (r_{max} - r_{min})) * 255$ 
#
# Mencari intensitas minimum dan maksimum secara manual

```

```

r_min = 255
r_max = 0

```

```

for y in range(height):
    for x in range(width):

```

```

        r = int(gray[y, x])

```

```

        if r < r_min:
            r_min = r

```

```

        if r > r_max:
            r_max = r

```

```

contrast = np.zeros((height, width), dtype=np.uint8)

```

```

if r_max == r_min:

```

```

    # Jika semua piksel memiliki nilai yang sama
    for y in range(height):
        for x in range(width):
            contrast[y, x] = gray[y, x]

```

```

else:

```

```

    for y in range(height):
        for x in range(width):

```

```

            r = int(gray[y, x])

```

```

            s = ((r - r_min) / (r_max - r_min)) * 255

```

```

            if s > 255:
                s = 255

```

```

            if s < 0:
                s = 0

```

```

            contrast[y, x] = int(s)

```

E. THRESHOLDING

```

# Threshold mengubah gambar menjadi hitam dan putih.
#
# Jika intensitas >= threshold → 255
# Jika intensitas < threshold → 0

threshold_value = 128

threshold = np.zeros((height, width), dtype=np.uint8)

for y in range(height):
    for x in range(width):

        r = int(gray[y, x])

        if r >= threshold_value:
            threshold[y, x] = 255
        else:
            threshold[y, x] = 0

# 3. HISTOGRAM MANUAL

# histogram[i] =
# jumlah piksel yang mempunyai intensitas i

histogram = [0] * 256

for y in range(height):
    for x in range(width):

        nilai = int(gray[y, x])

        histogram[nilai] += 1

# 4. CDF MANUAL

# CDF[i] =
# jumlah piksel dengan intensitas <= i

cdf = [0] * 256

cdf[0] = histogram[0]

for i in range(1, 256):

    cdf[i] = cdf[i - 1] + histogram[i]

# 5. CARI CDF MINIMUM

cdf_min = 0

for i in range(256):

    if cdf[i] != 0:

```

```

        cdf_min = cdf[i]
        break

# 6. HISTOGRAM EQUALIZATION

#
# Rumus:
#
#  $s = ((CDF(r) - CDF_{min}) / (N - CDF_{min})) * 255$ 
#

jumlah_pixel = height * width

mapping = [0] * 256

for i in range(256):

    if jumlah_pixel == cdf_min:

        mapping[i] = 0

    else:

        nilai_baru = (
            (cdf[i] - cdf_min)
            / (jumlah_pixel - cdf_min)
        ) * 255

        if nilai_baru > 255:
            nilai_baru = 255

        if nilai_baru < 0:
            nilai_baru = 0

        mapping[i] = int(nilai_baru)

# 7. TERAPKAN MAPPING KE GAMBAR

equalized = np.zeros((height, width), dtype=np.uint8)

for y in range(height):
    for x in range(width):

        nilai_lama = int(gray[y, x])

        nilai_baru = mapping[nilai_lama]

        equalized[y, x] = nilai_baru

# 8. TAMPILKAN HASIL

cv2.imshow("Gambar Asli", image)

cv2.imshow("Grayscale", gray)

```

```
cv2.imshow("Brightness", brightness)
cv2.imshow("Darkness", darkness)
cv2.imshow("Negative", negative)
cv2.imshow("Contrast_Stretching", contrast)
cv2.imshow("Thresholding", threshold)
cv2.imshow("Histogram_Equalization", equalized)
cv2.waitKey(0)
cv2.destroyAllWindows()
```